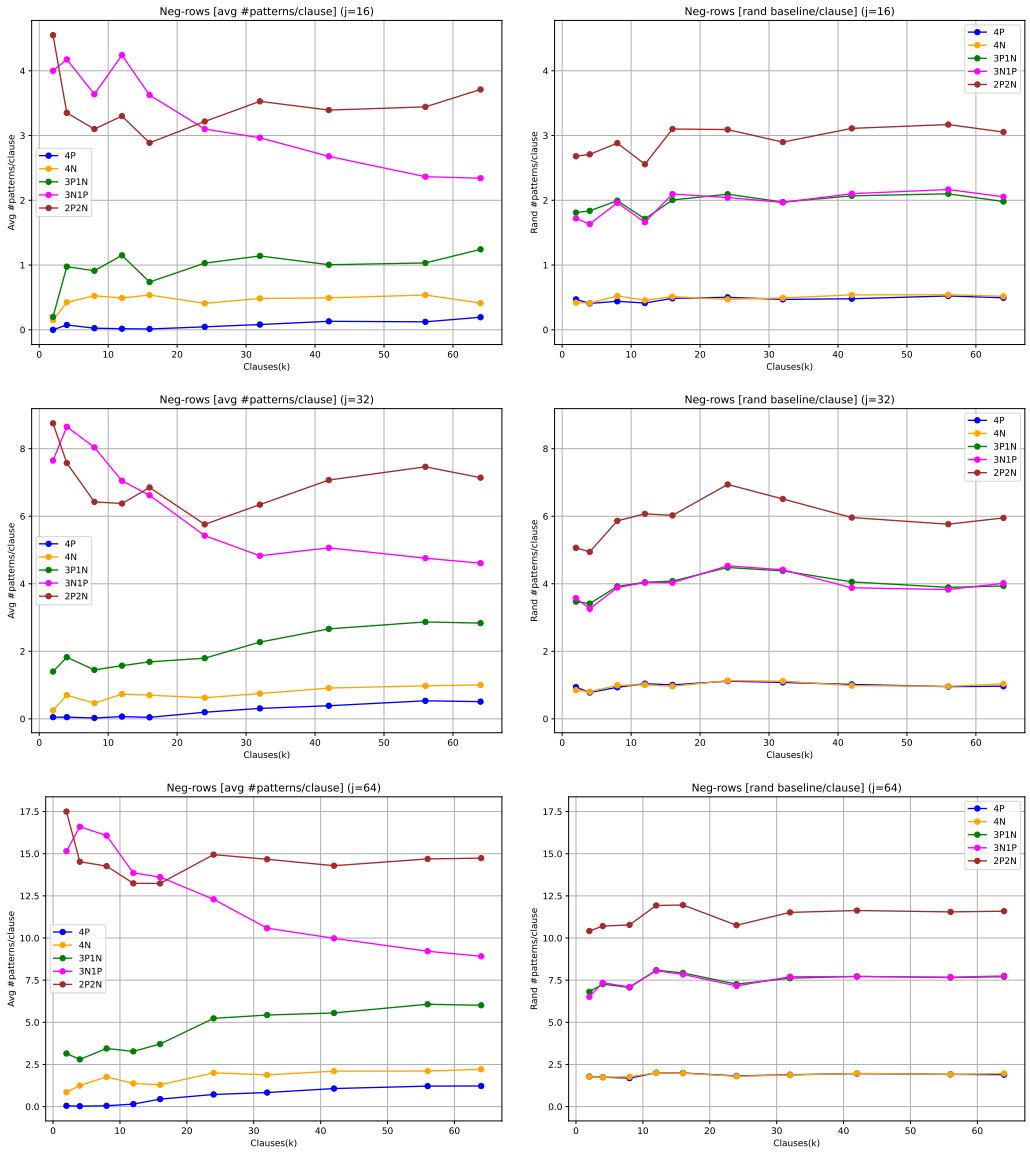

Figure 8: Prevalence of coding patterns in negative rows: trained networks versus random networks.

on the right side - the left side is a duplicate of Figure 4, shown again here for convenience. We see here that there is, in fact, a close alignment between training error and the number of clauses with no 4P coding rows. However, it does seem like the network can tolerate a small amount of clauses without 4P coding rows.

Next, we examine the overlap between the codes that the network is using. As shown in [1], this is a central aspect of coding. Codes usually overlap and always will if there is sufficient saturation

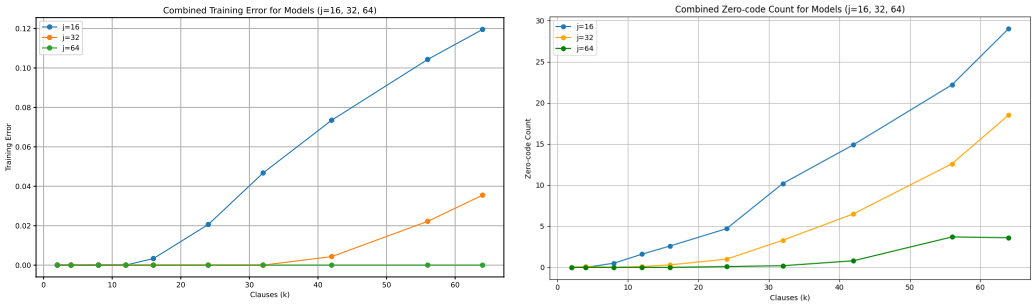

Figure 9: Scaling of training error and number of clauses without codes

of the network by the number of features being coded. However, as long as the overlap between pairs of codes is not too large, this is fine. One of the main advantages of using codes is that spreading out a signal across multiple neurons creates tolerance to noise that appears on some of those neurons. This is a core tenant of the feature channel coding hypothesis.

We show the code overlap measured from our experiments in Figure 10. We plot the average, over all pairs of clauses in each training run, of the overlap between the codes, where an overlap means that both codes use the same row as a 4P positive row. We also show the total number of 4P coding rows. For both values, we do not include clauses without coding rows in the count, so the blue curve (code size) will differ from that of Figure 7, which includes those clauses. These curves are almost identical (except for scaling) for the three hidden parameter sizes, and consistent with what we would expect from the feature channel coding hypothesis. Also, we do not see any indication that this overlap is responsible for the network not being able to train.

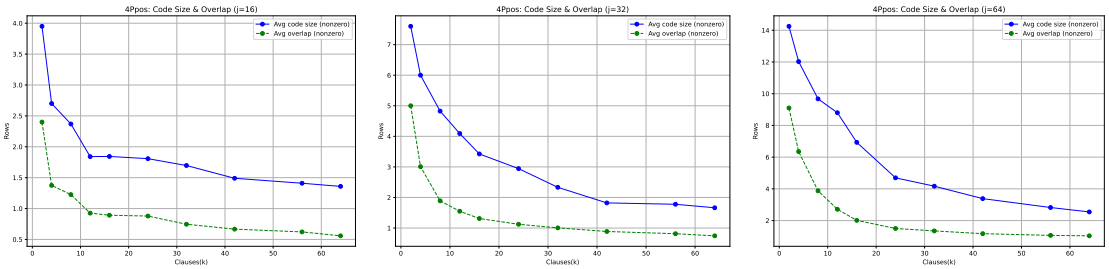

Figure 10: The average, over all pairs of clauses in each training run, of the overlap between the feature channel codes. Average code size is shown to give the reader a measure of the relative portion of the overlaps.

4.5 Towards Combinatorial Scaling Laws

We next show that the properties of feature channel coding explain the shape of traditional scaling laws. Unlike traditional scaling law graphs, our focus on Boolean formulas allows us to control the number of features explicitly, and we did so to construct Figure 7, where we studied the impact of increasing the number of features for three different sized networks. Here, in Figure 11, we show

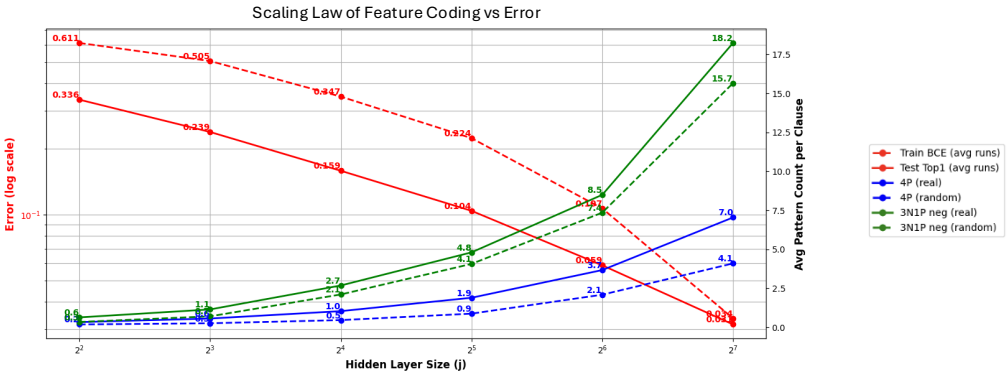

Figure 11: Feature channel coding combinatorially explains the neural network's scaling law.

similar data from a more traditional scaling law perspective, where we keep the number of features (clauses) fixed at $k = 64$, and increase the size of the hidden layer. We plot both the error and the frequency of 4P codes in positive rows (blue) and 3N1P codes in negative rows (green) versus the size of the hidden layer. We see that the decrease in error is closely related to the increase in code frequency. We believe that this relationship is due to the capacity of the network to perform feature channel coding. As discussed in Section 4.2, there is a limit to how much of the 4P pattern can fit into the network, and the network roughly achieves that limit.[1] This provides an easily explainable scaling law: the network's size implies a capacity to code the feature channels, with smaller networks being more limited in their coding ability, which explains the higher error rate.[2]

We also point out that in all cases, both patterns appear more frequently than random (solid lines versus dashed lines). The 4P positive coding is potentially more dominant as a code, hinted at by its relative ratio to the random distribution. As networks grow, the ratio of 4P to 4P random starts around 2, and decreases, but still is at almost 1.7x with 2^7 hidden neurons, while the ratio of 3N1P to 3N1P random is at 1.16 with 2^7 hidden neurons. One could envision building on this to use the ratios of codes in smaller models to derive scaling predictions for larger models, an approach that merits further research.

4.6 How Feature Channel Codes Emerge

We next turn to the question of how the network evolves using gradient descent. This is depicted in Figure 12. The results of this graph are interesting in their own right, but they also serve to again highlight the advantage of our combinatorial approach. We are able to gain insights into "what is learned when" during the training process, and these insights also serve to highlight what is important to the model learning process.

In the top part of Figure 12 we track the average number of patterns per clause as a network with $j = 32$ neurons and $k = 4$ clauses is trained through 20 epochs. To construct this graph,

[1] As before, when the hidden layer size increases and there is more space, the network starts to not need to take full advantage of that extra space.

[2] Note that we trained here to half the epochs of prior examples and there is a larger error per model size.